Decomposing Aggregate Productivity

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Abstract

In this note, we evaluate the sensitivity of commonly-used decompositions for aggregate productivity. Our analysis spans the universe of U.S. manufacturers from 1977 to 2012 and we find that, even holding the data and form of the production function fixed, results on aggregate productivity are extremely sensitive to how productivity at the firm level is measured. Even qualitative statements about the levels of aggregate productivity and the sign of the covariance between productivity and size are highly dependent on how production function parameters are estimated. Despite these difficulties, we uncover some consistent facts about productivity growth: (1) labor productivity is consistently higher and less error-prone than measures of multi-factor productivity; (2) most productivity growth comes from growth within firms, rather than from reallocation across firms; (3) what growth does come from reallocation appears to be driven by net entry, primarily from the exit of relatively less-productive firms.

Keyword*:* aggregate productivity, growth, misallocation, entry, exit

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1 Introduction

Aggregate productivity plays a central role in theories of economic growth in that it is the main source of long-run growth. This insight extends to even some neoclassical theories in business cycles. Despite its importance for our understanding of long-run economic growth and short-run aggregate fluctuations, surprisingly few robust facts on aggregate productivity (growth) have been documented in the literature. This is not only because of the scarce availability of micro-level data, but also due to different choices in measurement methodologies.

In this note, we highlight the sensitivity of commonly-used aggregate productivity decompositions by using confidential data covering the universe of U.S. manufacturers from 1977 to 2012. Similar to previous findings, we find that measures of aggregate productivity (growth) are extremely sensitive to how productivity is measured. First, we find that levels of aggregate productivity growth are substantially higher whenever firm-level productivity is measured through labor productivity, i.e. gross output per hours worked, rather than multi-factor productivity. This is not entirely surprising since, by construction, measures of labor productivity do not consider inputs such as capital, materials, and energy. As a result, aggregate labor productivity growth can reflect other mechanisms, such as capital deepening. What is surprising, however, is that aggregate labor productivity growth over the period 1977–2012 is *more than twice as high* when compared to multi-factor measures of aggregate productivity growth. The substantial difference between labor and multi-factor productivity could indicate either massive capital, materials, or energy deepening, or it could be the result of measurement error. Regardless of its source, we emphasize that these differences deserve further attention.

Second, we find that small differences in the way productivity is estimated at the firm level can have large consequences for our understanding of aggregate productivity, even when we hold the data and form of the production function fixed. To highlight this, we focus on two methods to estimate Cobb-Douglas production function parameters. Under

the first method, which is adopted by the U.S. Census Bureau, parameters for each input are determined by their respective industry-level cost shares. In the second method, we adopt the proxy variable estimator from De Loecker and Warzynski (2012) to estimate the production function parameters. Our main findings highlight three differences across these methods: (i) as in Foster, Haltiwanger and Krizan (2001), for any given period, the different measures of productivity disagree about the amount of overall productivity growth, often by several percentage points; (ii) the covariance between productivity and size is negative for most years when parameters are determined through industry-level cost shares, whereas this covariance is always positive when these parameters are estimated instead; (iii) the share of productivity growth from 1977–2012 coming from growth of the covariance between productivity and size varies widely across the three measures.

Despite the differences across measures, we do uncover some consistent patterns. First, in dynamic decompositions of productivity growth that account for entry and exit, most productivity growth comes from within-firm growth and the exit of unproductive firms. Reallocation across existing firms and entry of relatively more-productive usually accounts for only a small fraction of productivity growth. Second, periods of low overall productivity growth lead to large and offsetting values of the components, complicating inference. This problem is especially severe for measures of multi-factor productivity (which tend to be closer to zero than labor productivity) and for the productivity growth decomposition of Baily, Hulten and Campbell (1992), which contains two reallocation terms that offset each other.

The rest of this note is organized as follows. In section 2, we briefly review the literature on aggregate productivity decompositions and production function estimation. In section 3, we describe our measures of productivity and how we decompose them. Section 4 discusses our results, and section 5 concludes.

2 Literature Review

Our paper contributes mainly to the literature on the measurement and decomposition of aggregate productivity (growth). This includes the studies by Baily, Hulten and Campbell (1992), Griliches and Regev (1995), Bartelsman and Doms (2000), Aw, Chen and Roberts (2001), Foster, Haltiwanger and Krizan (2001), and Foster, Haltiwanger and Krizan (2006). The results in section 4 apply standard decompositions of aggregate productivity that have been used in several of the previously-mentioned studies. In particular, Foster, Haltiwanger and Krizan (2001) applied all three of our productivity decompositions to data on US manufacturing establishments (not firms) from 1977–1987 and found very different results across these decompositions. Furthermore, their results displayed substantial differences across (sub)industries. Bartelsman, Haltiwanger and Scarpetta (2013) show that across many countries in Europe the Olley-Pakes covariance between size and productivity is positive, growing, and increasing more rapidly in developing economies.

While there are substantial differences across these studies on the importance of different components for aggregate productivity growth, they do share some common insights: reallocation accounts for a non-negligible fraction of aggregate productivity growth and net entry seems especially important.

A large group of models in firm dynamics featuring producers with heterogeneous productivity is consistent with these common findings (e.g., Hopenhayn, 1992; Melitz, 2003; Luttmer, 2007). In particular, improvements in aggregate productivity are due to the reallocation of resources towards more productive firms. This can happen through the reallocation of resources among incumbent firms but also through the entry and exit of productive and less productive firms, respectively.

However, our findings indicate that the role of reallocation *between* firms is limited and often negative. In fact, we find that most productivity growth seems to come from growth *within* firms instead. While our results do not necessarily contradict canonical models of firm dynamics (since our results could be consistent with substantial reallocation across

plants *within the same firm*), it does imply that we should think more carefully about the boundaries of the firm. Most frameworks define firms simply as production technologies ruling out the notion of reallocation across plants within the same firm, although there are exceptions (Kehrig and Vincent, 2019).

Importantly, our findings have implications for our understanding of aggregate outcomes. For example, several studies have emphasized the role of reallocation across firms for economic growth (Acemoglu et al., 2013; Peters, 2020). Furthermore, our results can be used to discipline the role of misallocation for aggregate productivity in quantitative frameworks (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). In particular, a vast literature in finance studies the effects of financial frictions on aggregate productivity, with mixed results. On the one hand, Banerjee and Moll (2010) and Moll (2014) show that, so long as productivity shocks are persistent, productivity firms will eventually "outgrow" any financial frictions, and the effects of financial frictions on misallocation will be small, at least among continuing firms. On the other hand, Midrigan and Xu (2014) and Kochen (2021) calibrate dynamic models of firm growth, entry and exit subject to financial frictions and find that the main effects of financial frictions in Korea and a panel of European countries, respectively, on productivity growth are through entry and exit, rather than misallocation among continuing firms. Pancost (2017) calibrates a dynamic model to match the evolution of the Olley-Pakes decomposition among Indian manufacturing establishments and finds that growth over time in both terms of the decomposition can be explained by productivity growth within firms, rather than changes in financial frictions.

3 Methodology

The main dataset we use in this study is the Census of Manufactures (CMF), a quinquennial census of all manufacturing establishments in the United States undertaken by the U.S. Census Bureau. An advantage of the CMF is that it allows us to observe both real gross

output and inputs, including total labor hours, materials and energy inputs, and two types of capital (equipment and structures). To reduce the influence of outliers and measurement error, we winsorize all variables at the 1 percent level.

We conduct our analysis at the firm level using firm-level identifiers from the Longitudinal Business Database (LBD). This is because many establishments are linked together in multi-unit firms, and economic decisions are made at the firm rather than the establishment level. The LBD covers the universe of U.S. employer firms and is an unbalanced panel of both establishments and firms; by merging the LBD with the CMF, we are able to consider firm (as opposed to establishment) entry and exit, which is an important component of productivity growth in two of our decompositions. All of our productivity measures, described in more detail below, are computed at the establishment level; we then take weighted averages to aggregate them to the firm level for multi-unit firms.

3.1 Three Measures of Productivity

We use three measures of productivity. The first and simplest is labor productivity:

$$z_{ijt}^{L} \equiv \log\left(\frac{Y_{ijt}}{H_{ijt}}\right) \tag{1}$$

where Y_{ijt} and H_{ijt} are real gross output and total hours, respectively, at firm i in industry j at time t. Definition (1) has the benefit of simplicity, though because it ignores other inputs (particularly, capital) it may represent changes in the input mix, including capital deepening, rather than actual changes in productivity.

Second, we use two measures of multi-factor productivity (MFP). For ease of comparison, both measures assume a Cobb-Douglas production function and the same set of inputs: total hours H_{it}, equipment K_{it}^{E}, structures K_{it}^{S}, materials M_{it}, and energy E_{it}. The two measures differ in how they compute the Cobb-Douglas weights for each input: in the first MFP

measure, we use the standard industry-level NBER weights:

$$z_{ijt}^{\text{NBER}} \equiv \log Y_{ijt} - \alpha_{jt}^{L} \log\left(H_{ijt}\right) - \alpha_{jt}^{K^S} \log K_{ijt}^{S} - \alpha_{jt}^{K^E} \log K_{ijt}^{E} - \alpha_{jt}^{M} \log M_{ijt} - \alpha_{jt}^{E} \log E_{ijt} \tag{2}$$

where the α weights are the standard industry-level NBER cost shares of each input, averaged between t and $t-1$. In the second MFP measure, we estimate the Cobb-Douglas weights instead. While these weights are defined at the industry level, they are held fixed over time. In particular, we have:

$$z_{ijt}^{\text{DLW}} \equiv \log Y_{ijt} - \alpha_{j}^{L} \log\left(H_{ijt}\right) - \alpha_{j}^{K^S} \log K_{ijt}^{S} - \alpha_{j}^{K^E} \log K_{ijt}^{E} - \alpha_{j}^{M} \log M_{ijt} - \alpha_{j}^{E} \log E_{ijt}. \tag{3}$$

The α weights in equation (3) are estimated using the GMM-IV estimator of De Loecker and Warzynski (2012).[1]

3.2 Three Decompositions of Aggregate Productivity

A recent literature in macroeconomics has begun using establishment-level microdata to decompose aggregate productivity growth in order to better understand what drives economic growth.

In a seminal paper, Olley and Pakes (1996) decompose aggregate productivity into a simple-average term and a term reflecting the covariance between size and productivity:

$$Z_t = \sum_i w_{it} z_{it} = \underbrace{\frac{1}{N_t}\sum_{i=1}^{N_t} z_{it}}_{\text{average productivity } \overline{z}_t} + \underbrace{\sum_{i=1}^{N_t} (z_{it} - \overline{z}_t)\left(w_{it} - \frac{1}{N_t}\right)}_{\text{OP covariance}}, \tag{4}$$

[1] More details on this GMM-IV estimator can be found in De Loecker and Warzynski (2012) and Yeh, Macaluso and Hershbein (2022). The key moment conditions that identify the Cobb-Douglas parameters are based on inputs being determined in period $t-1$ being orthogonal to innovations to idiosyncratic productivity in period t.

where Z_t is log aggregate productivity at time t, w_{it} is the aggregate employment share of firm i and z_{it} is a measure of its productivity (for example, revenues per worker or multi-factor productivity). To ease notation, we omit industry subscripts j in this section. All decompositions are taken within industry, and later averaged across industries.

In addition to the Olley-Pakes decomposition given in equation (4), Foster, Haltiwanger and Krizan (2001) consider two dynamic decompositions of productivity growth, given by

$$\begin{aligned}\Delta Z_t &= \Delta \sum_i w_{it} z_{it} \\ &= \underbrace{\sum_{i \in C_t} w_{i,t-1} \Delta z_{it}}_{\text{within}} + \underbrace{\sum_{i \in C_t} \left(z_{i,t-1} - Z_{t-1}\right) \Delta w_{it}}_{\text{between}} + \underbrace{\sum_{i \in C_t} \Delta w_{it} \Delta z_{it}}_{\text{cross}} \\ &\quad + \underbrace{\sum_{i \in N_t} w_{it} \left(z_{it} - Z_{t-1}\right) - \sum_{i \in X_t} w_{i,t-1} \left(z_{i,t-1} - Z_{t-1}\right)}_{\text{net entry}},\end{aligned} \tag{5}$$

and

$$\begin{aligned}\Delta Z_t &= \Delta \sum_i w_{it} z_{it} \\ &= \underbrace{\sum_{i \in C_t} \overline{w}_i \Delta z_{it}}_{\text{within}} + \underbrace{\sum_{i \in C_t} \left(\overline{z}_i - \overline{Z}_t\right) \Delta w_{it}}_{\text{between}} \\ &\quad + \underbrace{\sum_{i \in N_t} w_{it} \left(z_{it} - \overline{Z}_t\right) - \sum_{i \in X_t} w_{i,t-1} \left(z_{i,t-1} - \overline{Z}_t\right)}_{\text{net entry}}\end{aligned} \tag{6}$$

where Z_t is aggregate productivity at time t, measured as a weighted average of plant- or firm-level productivities z_{it}, w_{it} is the associated weight of plant or firm i in aggregate productivity at time t, C_t is the set of firms that appear at both t and $t-1$ ("continuers"), N_t is the set of new plants or firms that appear only in t, and X_t is the set of plants or firms that exit (and thus appear only in $t-1$). A "Δ" next to variable denotes a difference between periods, while a bar over a variable denotes an average across periods. Both decom-

positions (5) and (6) characterize aggregate productivity growth in terms of effects coming from growth within firms holding their weights constant, growth across firms holding their productivities constant, and growth from net entry. In addition, decomposition (5) splits the "between" term into two pieces, one holding the weights constant and a second, "cross" term, reflecting the covariance between growth in weights and growth in productivity.

Decompositions (5) and (6), which are originally due to Baily, Hulten and Campbell (1992) and Griliches and Regev (1995), respectively, allow for a rich characterization of aggregate productivity growth into terms reflecting growth arising from changes within firms, across firms, and from the entry and exit of firms. However, these decompositions also require more detailed data than the OP decomposition (4); in particular, a panel dataset linking establishments over time. In addition, although decomposition (5) seems to offer a finer characterization of the "across" components into multiple terms, Foster, Haltiwanger and Krizan (2001) find these terms are unstable in practice and conjecture that they reflect more measurement error than anything else.

We apply decompositions (4), (5), and (6) to each 3-digit manufacturing NAICS code individually, and then report the weighted average, across industries, of each term. For the industry average, we use the same weighting variable as for w_{it} in each equation. For labor productivity, we use total hours H_{ijt} to compute weights, while for the two MFP measures we use real gross output Y_{ijt} to construct weights.

4 Results

This section reports the results of applying decompositions (4), (5), and (6) to the three measures of productivity described in Section 3.1. We find much disagreement between measures and decompositions: the covariance between size and productivity in equation (4) is positive for some measures and negative in others; moreover, its share of total productivity growth from 1977 to 2012 varies from just above 20 percent to over 100 percent, depending on the

measure. Although most labor productivity growth seems to come from within-firm growth and net entry in equations (5) and (6), it is difficult to reach similarly-strong conclusions about MFP in part because MFP growth is often small or negative. Decomposition (6) is easier to interpret and does seem to be less sensitive to measurement issues than (5).

4.1 Decomposition: Olley-Pakes

Table 1 reports the results of applying the Olley-Pakes decomposition (4) to the three measures of productivity defined in section 3.1. In the table, aggregate and average productivity are normalized to a value of 0 in 1977; thus the bottom row shows that while aggregate labor productivity grew by 69 percent from 1977 to 2012, MFP grew by either 31 percent (NBER measure) or by 12 percent (DLW measure). That aggregate productivity grew substantially more than MFP is not in itself surprising; the additional growth could merely reflect capital deepening. The differences between the two MFP measures—which use the same inputs and form of the production function—are more surprising.

The value of the Olley-Pakes covariance term varies widely across the three productivity measures, though it does seem to be growing over time. Surprisingly, and in contrast to the results in Foster, Haltiwanger and Krizan (2001), Bartelsman, Haltiwanger and Scarpetta (2013), and Pancost (2017), the Olley-Pakes covariance term is not even always positive: it is negative for six out of the eight years in our sample for the NBER-MFP measure, indicating that it is *smaller* firms (by gross output) which are more productive. The Olley-Pakes covariance is positive for the other two measures, and it is growing over time for all three.

Year	Labor Productivity			NBER-MFP			DLW-MFP		
	Overall	Within	Across	Overall	Within	Across	Overall	Within	Across
1977	0.00	-0.17	0.17	0.00	0.05	-0.05	0.00	-0.09	0.09
1982	-0.04	-0.20	0.15	0.07	0.09	-0.02	0.04	-0.03	0.07
1987	0.09	-0.08	0.17	0.12	0.14	-0.03	0.04	-0.05	0.09
1992	0.17	-0.06	0.22	0.09	0.15	-0.05	0.02	-0.08	0.10
1997	0.31	0.10	0.21	0.15	0.24	-0.10	0.03	-0.08	0.11
2002	0.47	0.25	0.22	0.24	0.27	-0.03	0.02	-0.12	0.14
2007	0.62	0.33	0.29	0.30	0.25	0.05	0.12	-0.08	0.20
2012	0.69	0.37	0.32	0.31	0.20	0.11	0.12	-0.10	0.22
Diff 1977–2012	0.69	0.54	0.15	0.31	0.15	0.16	0.12	-0.01	0.13
Share (%)	100	78	22	100	48	52	100	-8	108

Table 1. Olley-Pakes Decomposition, equation (4)
The table reports the results of the Olley-Pakes decomposition (4) for three measures of productivity: labor productivity from equation (1), NBER-MFP from equation (2), and DLW-MFP from equation (3). We compute all decompositions within each 3-digit NAICS manufacturing sector; the table reports weighted averages across industries, where the weights are total hours (for labor productivity) or gross output (for the two MFP measures). For ease of comparison, both the Overall and Within columns subtract the the value of overall productivity in 1977. The last two rows report the difference between the 2012 and 1977 values, and the ratio of that difference to overall productivity growth.

Although the Olley-Pakes covariance term contributes positively to productivity growth for all three measures, its share of overall growth is vastly different. For labor productivity, growth in the Olley-Pakes covariance represents only a fifth of labor productivity growth; for NBER-MFP growth it represents about half; and for DLW-MFP growth, the growing covariance between size and productivity accounts for more than 100 percent of growth from 1977 to 2012. The penultimate row of Table 1 reports the total change from 1977 to 2012 in each component for each productivity measure, and the bottom row expresses this value as a share of overall growth. Because the "within" term for DLW-MFP has actually shrunk over this period, the Olley-Pakes covariance term is responsible for more than 100 percent of the 12 percent productivity growth over this 36-year period. In contrast, the Olley-Pakes covariance accounts for 21 percent of the 69 percent growth in labor productivity, and 52

percent of the 31 percent growth in NBER-MFP.

The penultimate row of Table 1 shows that even though the *share* of growth over 1977–2012 coming from an increasing covariance between size and productivity varies across measures, the *amount* of growth accounted for by the covariance term is roughly the same (13–16 percentage points). Thus, one possibility is that methodological differences across the three measures—for example, capital deepening leading to higher observed labor productivity growth, or the different identification assumptions between NBER-TFP and DLW-TFP—tend to load on within-firm productivity, and that in fact the amount of productivity growth from reallocation across firms is roughly the same no matter how productivity is measured. If true, such a conclusion would be unique to the Olley-Pakes decomposition, as we show in the next two subsections (where no such agreement between measures exists).

4.2 Decomposition: BHC

Table 2 reports the results of applying the dynamic decomposition (5) to our three measures of productivity. Each row reports growth rates for different 5-year periods, and each panel reports a separate measure of productivity. The column labeled "Overall" reports aggregate productivity growth. As in Table 1, labor productivity growth far outstrips MFP growth over the entire sample period, being less than 14 percent in only three five-year periods, while MFP growth (either measure) is never higher than 13 percent. In fact, MFP growth is often close to zero or even negative. Moreover, years of outsize productivity growth in MFP do not align across measures: NBER-MFP grew by almost 10 percent from 1997 to 2002, when DLW-MFP grew by less than 1 percent; on the other hand, DLW-MFP growth was more than twice as high in 2002–2007 than NBER-MFP growth.

Year	Overall	Within	Between	Cross	Entry	Exit
Labor Productivity						
1977–1982	-0.92	4.45	0.40	-4.14	-3.56	1.85
1982–1987	15.07	0.75	0.15	-0.30	0.30	0.11
1987–1992	6.10	0.79	0.46	-0.81	0.08	0.48
1992–1997	15.29	0.95	0.12	-0.36	0.22	0.07
1997–2002	14.38	0.78	0.20	-0.32	0.24	0.10
2002–2007	14.22	0.85	0.24	-0.41	0.18	0.14
2007–2012	4.71	1.24	0.67	-1.19	-0.24	0.53
NBER-MFP						
1977–1982	3.65	0.54	-0.29	0.40	0.20	0.15
1982–1987	2.02	0.76	-0.49	0.70	-0.18	0.21
1987–1992	-3.42	-0.65	-0.26	0.30	-0.58	0.19
1992–1997	1.33	0.10	-1.02	1.78	0.06	0.09
1997–2002	9.79	0.29	-0.07	0.50	0.30	-0.03
2002–2007	5.25	-0.30	-0.08	0.85	0.34	0.19
2007–2012	0.54	-7.99	-0.40	8.09	0.97	0.33
DLW-MFP						
1977–1982	4.03	0.11	-0.24	0.53	0.41	0.19
1982–1987	0.49	-1.26	-1.55	3.81	-0.53	0.53
1987–1992	-0.84	-1.48	-0.91	1.85	-1.03	0.57
1992–1997	3.26	0.37	-0.40	0.83	0.12	0.07
1997–2002	0.61	-3.37	-1.10	5.52	-0.40	0.36
2002–2007	12.84	0.52	-0.08	0.34	0.21	0.01
2007–2012	0.53	-4.74	-3.03	8.49	-0.17	0.44

Table 2. Dynamic Decomposition BHC, equation (5)
The table reports the results of the dynamic decomposition (5) for three measures of productivity: labor productivity from equation (1), NBER-MFP from equation (2), and DLW-MFP from equation (3). We compute all decompositions within each 3-digit NAICS manufacturing sector; the table reports weighted averages across industries, where the weights are total hours (for labor productivity) or gross output (for the two MFP measures). The "Overall" column reports aggregate productivity growth between the years indicated in the first column; subsequent columns report the share of overall productivity growth attributable to each component of aggregate productivity.

The last four columns of Table 2 report the shares of productivity growth attributable to each component of the decomposition (5): within, between, "cross," entry, and exit.

For labor productivity, the vast majority of productivity growth comes from within-firm productivity growth: in all years, the within term accounts for more than 75 percent of labor productivity growth. This result lines up well with the fact from Table 1 that 78 percent of overall productivity growth from 1977 to 2012 came from the within component.

Before further analyzing the components of productivity growth in Table 2, we note that in years where overall productivity growth is small or negative, the shares tend to swing wildly across columns; for example, from 1977–1982 when labor productivity growth was roughly -1 percent, over 400 percent of the drop was from the cross term, another 350 percent of the drop was from entry, and the overall effect was only -0.92 percent because of the large positive contributions of the within and exit terms.[2] A similar pattern appears in the two MFP terms. Thus, we focus the rest of our analysis on years when productivity growth is far from zero.

Although within-firm productivity growth accounts for more than 75 percent of labor productivity growth, there is no similarly-clean conclusion about within-firm growth from the two MFP measures. Ignoring the two periods of smallest productivity growth for each measure, the within component ranges from 29–76 percent of NBER-MFP growth, and from 11–337 percent of DLW-MFP growth. Even when restricting attention to periods when overall MFP growth is high, it represents only 29–30 percent of NBER-MFP growth (1997–2007) and 11–52 percent of DLW-MFP growth (1977–1982 and 2002–2007).

Decomposition (5) contains two terms relating to reallocation across continuing firms ("between" and "cross"), which are of opposite sign in all years and for all measures of productivity. For labor productivity, the "cross" terms are all negative and larger in magnitude than the "between" terms, suggesting a negative effect of reallocation, while the reverse is true for the two MFP measures. Moreover the magnitudes are large; to see this more clearly, Table 3 sums the two reallocation terms from Table 2.

[2]Note that we define shares as component growth rates divided by the absolute value of overall growth, so that when overall growth is negative shares sum to -1, and the sign of the share reflects the sign of the effect on growth.

Table 3 shows that while reallocation among continuers appears to be a consistent drag on labor productivity, reducing labor productivity growth by about 15–30 percent, it tends to *increase* MFP growth, sometimes accounting for 30–40 percent of MFP growth. Notice that even though Table 3 omits for each measure of productivity the two years when overall productivity growth was closest to zero, there are nevertheless years when reallocation seemingly contributed 77 percent of NBER-MFP growth, or 442 percent of DLW-MFP growth. We view these estimates as driven by outliers; labor productivity, being a simpler concept with fewer ingredients, seems less prone to the effect of outliers.

Year	Labor	NBER-MFP	DLW-MFP
1977–1982		0.11	0.29
1982–1987	-0.15	0.21	
1987–1992	-0.35	0.05	0.94
1992–1997	-0.25		0.44
1997–2002	-0.12	0.43	4.42
2002–2007	-0.17	0.77	0.25
2007–2012			

Table 3. Sum of Reallocation Terms in (5)
The table reports the sum of the Between and Cross columns from Table 2, for the three measures of productivity, omitting (for each measure of productivity) the two years in which overall productivity growth was closest to zero.

At first glance, the negative reallocation term for labor productivity in Table 3 seems to contradict the fact that the Olley-Pakes covariance term for labor productivity in Table 1 is positive, growing, and responsible for about 21 percent of overall productivity growth from 1977–2012. In fact, there is no contradiction; the difference stems from the fact that decomposition (5) explicitly accounts for net entry, while the Olley-Pakes decomposition (4) does not. The last two columns of Table 2 show that net entry has a strong positive effect on labor productivity growth in all years apart from 1977–1982. Moreover, in most years (in particular years when productivity growth is far from zero) the combined positive effects of entry and exit are larger than the combined effects of the cross and between terms, as can

be seen in Table 4.

Year	Reallocation	Net Entry	Sum
1977–1982			
1982–1987	-0.15	0.41	0.25
1987–1992	-0.35	0.56	0.21
1992–1997	-0.25	0.29	0.05
1997–2002	-0.12	0.34	0.22
2002–2007	-0.17	0.33	0.15
2007–2012			

Table 4. Labor Productivity: Reallocation vs. Net Entry
The table reports the sum of the Between and Cross terms as well as the sum of the entry and exit terms from decomposition (5), for labor productivity, from Table 2, omitting the two years in which overall labor productivity growth was closest to zero.

Comparing the effects of the two net entry terms and the two reallocation terms in Table 4 with the growth of the Olley-Pakes covariance term for labor productivity in Table 1 suggests that the Olley-Pakes covariance growth is driven mainly by net entry. Table 4 shows that, in years when labor productivity growth is far from zero, the combined effect of reallocation and net entry is generally between 15–20 percent. The positive effect on productivity growth is entirely driven by net entry, and *despite* the negative contribution of reallocation among continuing firms. As for which component of net entry matters more for growth, it appears to change over time; both entry of more-productive and exit of less-productive firms seem to increase productivity growth, with entry more important in some years (1982–1987, 1992–2002) and exit in others (1987–1992).

4.3 Decomposition: GR

Foster, Haltiwanger and Krizan (2001) note that decomposition (5) might be more prone to measurement errors, and suggest the alternative (6), which is closely related to a decomposition suggested by Griliches and Regev (1995). Indeed, the many extreme values in Table 2 make inference difficult, and the distinction between the between and cross terms

seems particularly unhelpful if they are consistently of opposite sign. Table 5 reports the results of applying the dynamic decomposition (6) to our three measures of productivity. As in Table 2, each row reports growth rates for a different pair of years, each panel reports a separate measure of productivity, the column labeled "Overall" reports aggregate productivity growth (identical to Table 2), and subsequent columns report the share of aggregate productivity growth attributable to each component from equation (6).

Year	Overall	Within	Between	Entry	Exit
Labor Productivity					
1977–1982	-0.92	2.38	-1.69	-3.31	1.62
1982–1987	15.07	0.59	-0.01	0.11	0.30
1987–1992	6.10	0.39	0.06	-0.12	0.68
1992–1997	15.29	0.77	-0.05	0.06	0.23
1997–2002	14.38	0.62	0.00	0.08	0.31
2002–2007	14.22	0.64	0.01	0.05	0.31
2007–2012	4.71	0.64	0.04	-0.36	0.68
NBER-MFP					
1977–1982	3.65	0.74	-0.09	0.06	0.29
1982–1987	2.02	1.11	-0.17	-0.30	0.35
1987–1992	-3.42	-0.50	-0.12	-0.42	0.04
1992–1997	1.33	0.99	-0.18	-0.03	0.22
1997–2002	9.79	0.54	0.13	0.15	0.18
2002–2007	5.25	0.12	0.28	0.18	0.42
2007–2012	0.54	-3.95	3.16	0.27	1.51
DLW-MFP					
1977–1982	4.03	0.37	0.03	0.26	0.35
1982–1987	0.49	0.65	0.39	-0.65	0.62
1987–1992	-0.84	-0.56	-0.03	-0.85	0.43
1992–1997	3.26	0.79	0.02	-0.03	0.22
1997–2002	0.61	-0.61	1.61	-0.49	0.49
2002–2007	12.84	0.69	0.05	0.09	0.17
2007–2012	0.53	-0.49	0.88	-0.36	0.97

Table 5. Dynamic Decomposition GR, equation (6)
The table reports the results of the dynamic decomposition (6) for three measures of productivity: labor productivity from equation (1), NBER-MFP from equation (2), and DLW-MFP from equation (3). We compute all decompositions within each 3-digit NAICS manufacturing sector; the table reports weighted averages across industries, where the weights are total hours (for labor productivity) or gross output (for the two MFP measures). The "Overall" column reports aggregate productivity growth between the years indicated in the first column; subsequent columns report the share of overall productivity growth attributable to each component of aggregate productivity.

Although years of low aggregate productivity growth (such as 1977–1982 for labor productivity, or 2077–2012 for both measures of MFP) still tend to imply implausibly-large share

values for the components, the swings are noticeably smaller than in Table 2. For example, the within component for labor productivity is 2.38 times the change in productivity from 1977–1982, rather than 4.45 times. Likewise, component values for both MFP measures from 2007–2012 are similarly muted relative to Table 2. Thus, we view this decomposition as more robust than decomposition (5), though we still focus most of our attention on years of productivity growth that are far from zero since a few wild swings remain.

Table 5 shows that most labor productivity growth comes from the within-firm component (between 60 and 77 percent) and the exit component (between 23 and 68 percent). In contrast, the between component is small or even negative, generally not more than 6 percent of productivity growth. Likewise, entry seems to have only a modest effect on labor productivity growth, accounting for at most 11 percent in 1982–1987. Thus unlike in decomposition (5), where the reallocation component generally has a negative effect on productivity growth which is compensated by a large effect from net entry (Table 3), here the reallocation effect is negligible but the net entry effect, particularly from exit, is strong.

Likewise, the within and exit components of the two MFP measures are the biggest contributors to growth, and the between component is generally quite small—though it is a bit harder to see than for labor productivity growth. Omitting years of overall MFP growth close to zero, the within component contributes to between 50 and 111 percent of NBER-MFP growth (apart from 12 percent in 2002–2007) and between 49 and 79 percent of DLW-MFP growth (apart from 37 percent in 1977–1982). The exit component accounts for between 18 and 42 percent of NBER-MFP growth (apart from 1987–1992 when it was positive 4 percent when overall productivity fell) and between 17 and 49 percent of DLW-MFP growth (apart from, again, 1987–1992 when it was positive 43 percent when overall productivity fell). The entry component is often the opposite sign from exit for NBER-MFP and occasionally quite large (-30 percent of growth in 1982–1987, and 15–18% of growth in 1997–2007). Apart from its 26 percent contribution to DLW-MFP growth in 1977–1982, entry does not contribute much to DLW-MFP growth outside of years with low overall

growth. The between component for NBER-MFP is somewhat larger than the other two measures: it reaches 28 percent in 2002-2007 (when the within component is so small) and is 13 percent of growth in 1997–2002. For DLW-MFP, the between component is only larger in periods when overall MFP growth is close to zero.

5 Conclusion

We construct and decompose a long time-series of U.S. manufacturing productivity and find that results are sensitive to methodological details of how productivity is measured, holding fixed the data and form of the production function. While some results emerge clearly from our analysis—labor productivity growth is an order of magnitude higher than MFP growth, and is mostly driven by within-firm growth and net entry—other results vary across measures and decompositions, including how much growth comes from within, between, and net entry components, and when productivity growth is high or low. When productivity growth is close to zero, the shares of productivity growth from individual components tend to take implausibly-large and offsetting values.

The obvious question, which we are unable to answer, is *why* these different measures and decompositions yield such drastically different conclusions from one another. Although the differences between labor productivity and MFP might reflect important economic phenomena—such as capital deepening—the magnitude of the difference is large enough that additional investigation is needed. Likewise, the differences between the two MFP measures—whose main but not only methodological difference is the time-varying vs. constant input weights—also merit further analysis.

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